

The ecomania Value System

A five-axis model of ecological worldview for the ecoavatar.

Abstract

ecomania positions each user in a shared *value-axis space* derived from an onboarding survey, snaps them to the nearest **archetype** (the avatar's base form), and thereafter nudges a **live position** via topical questionnaires. This requires a small set of axes that (a) are grounded in validated environmental-psychology instruments, (b) are mutually near-independent so the avatar space does not collapse, and (c) each map cleanly to one *non-overlapping* visual register that can drive a Rive state machine. We review ~15 frameworks and propose **five axes**: *Anthropocentric-Ecocentric*, *Technological-Sufficiency*, *Individual-Collective*, *Technocratic-Grassroots*, and *Local-Global*. We give each axis a definition, the policy debate it captures, its empirical grounding, an independence argument, and its avatar channel; we then analyse residual collinearity and the design implications.

1. Motivation: why a *small, deliberately chosen* axis set

The most rigorously factor-analysed instrument in the field, the Environmental Attitudes Inventory ([Milfont & Duckitt 2010](#)), collapses **12** attitude facets into just **two** correlated higher-order factors — *Preservation* and *Utilization*. The New Ecological Paradigm scale ([Dunlap et al. 2000](#)) is treated by its authors as essentially a **single** general "ecological worldview" dimension. The lesson is that environmental ideology is **low-dimensional**: naively chosen axes will each be a facet of "how green are you" and the population will collapse onto a diagonal, making archetypes indistinguishable.

This is especially acute for ecomania, whose users self-select into an ecology app and are therefore already green — the general-greenness dimension has little variance among them. The design discipline that follows is:

- spend axes on **what kind of green** (orientation → archetype + surface), and
- let **how much green** (intensity/urgency) live on a separate **engagement clock** — the segmentation captured by Global Warming's Six Americas ([Leiserowitz et al. 2009](#)), which is a single Alarmed→Dismissive engagement gradient, *not* a values space.

Each axis below is therefore an *orientation* axis, and each is assigned to a distinct visual channel. Mutual visual non-overlap is the practical test that the axes are well-separated: a designer can move any one Rive input without the others visually contradicting it.

2. Approach

This is a *synthesis*, not new empirical work. Axes were selected by triangulating recurring dimensions across value theory ([De Groot & Steg 2008](#)), attitude structure ([Milfont & Duckitt 2010](#); [Dunlap et al. 2000](#)), political-theory discourse typologies ([Dryzek 2013](#)), cultural theory of risk ([Douglas & Wildavsky 1982](#); [Kahan et al. 2011](#)), and environmental-spatial cognition ([Uzzell 2000](#)). Where multiple independent frameworks "rediscover" the same dimension, we treat it as robust. Independence claims are qualitative (directional), drawn from the cited studies, not from a new factor analysis.

3. The five axes

Axis 1 — Anthropocentric ↔ Ecocentric

Whose interests have moral standing?

– pole (Anthropocentric)	Nature matters for human benefit — health, food, economy, recreation, future human generations.
+ pole (Ecocentric)	Nature has intrinsic value; species, rivers, ecosystems warrant moral consideration in their own right.
Policy debate	Rights-of-nature law and intrinsic-value conservation vs ecosystem-services / natural-capital accounting.
Contested case	Spain's Mar Menor lagoon became Europe's first ecosystem with legal personhood (Law 19/2022, born of a 640,000-signature citizen initiative; upheld by the Constitutional Court in Nov 2024). <i>Ecocentric</i> reading: the lagoon holds intrinsic rights "to exist, evolve, be protected and restored." <i>Anthropocentric</i> counter: the farming, tourism and property interests whose nutrient run-off had collapsed it. (Sister case: New Zealand's Whanganui River, 2017.)
Grounding	O'Riordan 1976 's master ecocentric-technocentric spectrum; the biospheric-vs-altruistic value split of De Groot & Steg 2008 ; the EAI "ecocentric concern" vs "human utilization" facets (Milfont & Duckitt 2010); NEP anti-anthropocentrism (Dunlap et al. 2000).
Independence	A <i>value</i> about ends, silent on <i>means</i> — orthogonal to Axes 2–5. Thompson & Barton 1994 show ecocentric and anthropocentric motives predict behaviour independently.
Avatar channel	Silhouette / kingdom — humanoid/built/geometric ↔ wild/animal/plant/fungal.

Note (Limitation L1). [Thompson & Barton 1994](#) find ecocentrism and anthropocentrism are technically *two* dimensions, not opposite poles (one can score high or low on both). We model them as one bipolar axis; the degenerate "low-low" corner (cares about neither) is captured instead by the engagement clock (disengagement), and "high-high" sits at the axis midpoint.

Axis 2 — Technological ↔ Sufficiency

How do we reconcile prosperity with the planet?

– pole (Technological)	Innovation and growth can <i>decouple</i> wellbeing from harm: nuclear, renewables, lab-grown food, dense cities, carbon capture.
+ pole (Sufficiency)	Hard biophysical limits exist; efficiency rebounds (Jevons). The answer is to consume less — degrowth, post-growth, voluntary simplicity.
Policy debate	Ecomodernism (Asafu-Adjaye et al. 2015) vs degrowth (D'Alisa et al. 2014); the "bright green" vs "dark green" split.
Contested case	Nuclear power. Germany shut its last three reactors on 15 April 2023 , while the EU labelled nuclear "green" in its investment taxonomy (Parliament declined to block it, 328–278, July 2022). <i>Technological/ecomodernist</i> : nuclear is essential clean baseload. <i>Sufficiency/dark-green</i> : a costly distraction from consuming less

	(Greenpeace sued, alleging "greenwashing"). A second front: Italy (2023) and Florida (2024) banned cultivated meat — engineered-food optimism vs precaution/tradition.
Grounding	The ecomodernist/degrowth literatures above; the EAI "confidence in science and technology" facet (Milfont & Duckitt 2010).
Independence	A <i>means</i> question, orthogonal to Axis 1 (one can be an ecocentric technologist or an anthropocentric degrowther). Its main collinearity is with the <i>posture</i> construct (reformist↔radical), which we deliberately route to the engagement clock rather than making it an axis (see §4).
Avatar channel	Material / texture — sleek/metallic/luminous ↔ earthen/woven/mineral/raw.

Axis 3 — Individual ↔ Collective (*the "group" dimension*)

Where does change come from?

– pole (Individual)	Change is the sum of personal choices and market signals; responsibility is personal (recycle, fly less, buy an EV).
+ pole (Collective)	Individual action is marginal against systemic forces; what matters is policy, regulation and organising.
Policy debate	The carbon-footprint / consumer-responsibility framing vs "system change, not climate change" — the individualization critique of Maniates 2001 .
Contested case	The " carbon footprint " itself: popularised by BP's 2004-06 Ogilvy & Mather campaign (and personal footprint calculator) to frame climate as a matter of individual choice — later widely criticised as deflecting from systemic/corporate responsibility, making Maniates 2001 's thesis concrete. <i>Individual</i> camp: shrink your footprint, fly less ("flygskam"). <i>Collective</i> camp: the responsibility lies with ~100 producers and the policy that enables them.
Grounding	Maniates 2001 ; Stern 2000 's empirically distinct private-sphere, public-sphere and activism behaviour types; the individualism–communitarianism dimension of Kahan et al. 2011 .
Independence	Strong. Lacroix et al. 2022 find near-zero spillover from private green behaviour to collective action — so this axis separates people a single greenness score would merge.
Avatar channel	Multiplicity — a lone, self-contained creature ↔ accompanied by a flock/colony/mycelial network.

Axis 4 — Technocratic ↔ Grassroots (*the "grid" dimension*)

Who should be in charge of solving it?

– pole (Technocratic /	Trust experts, institutions and scale: carbon markets, IPCC-led policy, mega-projects, professional management.

hierarchical)	
+ pole (Grassroots / egalitarian)	Trust distributed community power: energy co-ops, indigenous stewardship, commons, citizens' assemblies.
Policy debate	Top-down carbon pricing and large infrastructure vs community-owned renewables and community/indigenous-led conservation.
Contested case	France's Citizens' Convention for Climate (150 randomly-selected citizens; 149 proposals, 2019–2020). Macron pledged to pass them "unfiltered," but the 2021 Climate & Resilience law kept only ~10 intact and watered down the rest — prompting a public revolt by convention members. <i>Grassroots/egalitarian</i> : deliberative citizen assemblies should set climate policy directly. <i>Technocratic</i> : expert ministries and Parliament must filter, cost and reconcile proposals.
Grounding	The hierarchy-egalitarianism dimension of cultural cognition (Kahan et al. 2011); the "grid" axis of cultural theory (Douglas & Wildavsky 1982 ; Steg & Sievers 2000).
Independence	High, and <i>distinct from Axis 3</i> : Axis 3 asks individual-vs-collective; Axis 4 asks, given collective action, top-down-vs-bottom-up. Cultural cognition treats grid and group as roughly orthogonal. The off-diagonal cases are real: <i>global-grassroots</i> (transnational movements) and <i>local-technocratic</i> (municipal expert planning).
Bonus	Axes 3 × 4 (group × grid) are exactly the plane that generates cultural theory's four "myths of nature" (Steg & Sievers 2000) — providing validated archetype anchors (Individualist/Nature-Benign, Egalitarian/Nature-Ephemeral, Hierarchist/Nature-Perverse-Tolerant, Fatalist/Nature-Capricious).
Avatar channel	Structure / form-order — symmetric/architected/crystalline ↔ asymmetric/organic-cluster/rhizomatic.

Note (Limitation L2). Cultural cognition **replicates only weakly outside the US/Anglo context** ([Pröpper et al. 2022](#), N = 44,378 across 23 countries). In non-Anglo markets this axis may carry less signal than the others.

Axis 5 — Local ↔ Global (*the spatial / scale dimension*)

At what scale do you locate the problem and the solution?

– pole (Local)	The watershed, neighbourhood, bioregion; place-based stewardship; "act local".
+ pole (Global)	The planet, global systems, international governance; "think planetary".
Policy debate	Bioregional / localist environmentalism vs cosmopolitan, treaty- and systems-scale framing.

Contested case	Serbia's Jadar lithium project (Rio Tinto, EU-backed). The licence — revoked after 2022 protests — was reinstated in July 2024, bringing tens of thousands onto the streets. <i>Global</i> : Europe needs the lithium for EV batteries and the energy transition. <i>Local</i> : the mine threatens the Jadar valley's farmland and water. The textbook local-place vs planetary-system clash — and note it cuts across Axis 4 (the protests were grassroots, the project EU-technocratic).
Grounding	Spatial cognition of environmental problems: "environmental hyperopia" — distant problems judged more serious than local ones (Uzzell 2000); the <i>spatial-optimism bias</i> across 18 nations (Gifford et al. 2009); place attachment and local-vs-global framing of engagement (Scannell & Gifford 2013).
Independence	Moderate (see §4, Limitation L3): it can correlate with both Axis 3 and Axis 4. Retained because it captures a distinct <i>scale-of-concern</i> the others do not, and grounds the "global systems thinker" identity.
Avatar channel	Scale / rootedness — rooted, place-bound (tree, soil) ↔ migratory, planetary (ocean-current, migratory bird).

4. Independence and residual collinearity

The five axes were chosen to maximise mutual independence, but the literature is explicit that environmental ideology is low-dimensional, so residual correlation is expected and *managed* rather than eliminated.

Axis pair	Expected collinearity	Why / mitigation
1 × (2,3,4,5)	Low	A value about ends; orthogonal to means/agency/governance/scale (Thompson & Barton 1994).
2 × 4	Low-moderate	Hierarchical-individualists discount environmental risk (Kahan et al. 2011); not deterministic (ecomodernism is often <i>communitarian-technocratic</i> , e.g. state-led nuclear).
3 × 4	Low	Validated as roughly orthogonal (group vs grid); off-diagonal cells populated.
3 × 5	Moderate (flag)	"Local" can read as community/individual scale, "global" as systemic/collective.
4 × 5	Moderate (flag)	"Grassroots" skews local; "technocratic" skews global.
(2,posture)	High — handled by design	Reformist↔radical bundles with tech↔sufficiency (Dryzek 2013 ; bright/dark green). Posture is an <i>intensity</i> construct, so it is assigned to the engagement clock , not an axis — this is why the model has 5 axes and not 6.

Design consequence. Axes 1–4 are treated as the *primary* identity basis (high independence, each backed by multiple frameworks). Axis 5 is *secondary*: included for thematic richness and the global-

thinker identity, but with the lowest empirical weight and the most overlap, so scoring should down-weight it relative to Axes 1–4, and it is the first candidate to drop if data shows it is redundant.

5. Design implications

- **Archetypes.** Use the Axis 3 × Axis 4 plane (group × grid) to anchor a validated 4-archetype core (the myths of nature), then differentiate base forms further with Axes 1–2; Axis 5 modulates surface/flavour rather than defining new base forms. (Each base form = one .riv ; archetype count = designer cost.)
- **Two clocks.** Orientation (Axes 1–5) → archetype + surface. Intensity/engagement (Six Americas gradient; reformist→radical posture) → leveling, aura, charge — the thing gameplay grows over time.
- **Visual contract.** Five non-overlapping channels — silhouette, material, multiplicity, structure, scale — plus an intensity/aura channel from the clock. Each is a numeric Rive input that can be driven independently.
- **GDPR / integrity.** Ecological values are special-category data. Survey item labels must be **balanced in valence** across both poles (e.g. avoid pairing a pejorative "technocratic" with a warm "communitarian"); neutral pole labels such as *Institutional* ↔ *Community* or *Vertical* ↔ *Horizontal* are preferred to non-leading the respondent.

6. Limitations

- **L1 — Axis 1 bipolarity.** Modelled as one axis though [Thompson & Barton 1994](#) treat ecocentrism/anthropocentrism as two dimensions (§3, Axis 1).
- **L2 — Axis 4 cross-cultural validity.** Grid replicates weakly outside Anglo contexts ([Pröpper et al. 2022](#)); weigh against early non-Anglo market plans.
- **L3 — Axis 5 weakness.** Thinnest grounding and highest overlap (with Axes 3 and 4); flagged as secondary and droppable.
- **L4 — Synthesis, not measurement.** Independence is argued from prior studies, not a new factor analysis on the ecomania population. A confirmatory factor analysis on early survey data should validate the axis structure and re-estimate the inter-axis correlations.

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Contested cases (news & policy)

- **Mar Menor** (Axis 1): Heinrich Böll Stiftung (2025), *Mar Menor: Europe's First Ecosystem with Legal Personhood* — Spain Law 19/2022; Constitutional Court upheld it Nov 2024. <https://www.boell.de/en/2025/02/05/mar-menor-europes-first-ecosystem-legal-personhood>
- **Nuclear phase-out** (Axis 2): NPR (2023), *Germany Begins Powering Down Its Last Three Nuclear Plants*. <https://www.npr.org/2023/04/15/1170244609/germany-begins-powering-down-nuclear-plants> — and European Parliament (2022), *Taxonomy: MEPs do not object to inclusion of gas and nuclear activities*. <https://www.europarl.europa.eu/news/en/press-room/20220701IPR34365/taxonomy-meps-do-not-object-to-inclusion-of-gas-and-nuclear-activities>
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